

Task 1 – WPA2 - Enterprise / RADIUS Capability Test

What was tested:

A Packet Tracer test was done using a Cisco WLC-2504, Lightweight AP-3702i, RADIUS server, and wireless laptop to test WPA2-Enterprise/RADIUS authentication.

Expected vs. actual result:

The result was for the AP to join the WLC and Laptop0 to authenticate through RADIUS. The AP successfully joined the WLC, and Laptop0 also successfully connected to the wireless network using the RADIUS account.

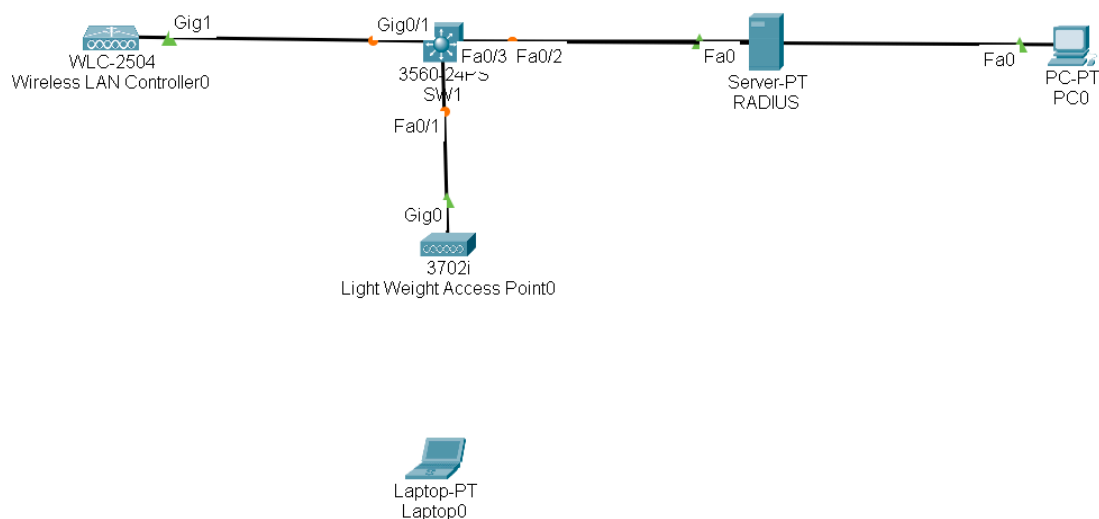
Result:

The test was successful. Packet Tracer does support WPA2-Enterprise/RADIUS authentication using the WLC-2504 and Lightweight AP-3702i

Limitation/consideration:

The WLC-2504 used RADIUS authentication port 1812 and the Server-PT used port 1645. Therefore, the server port had to be changed to 1812 to match the WLC. After this, the WPA2-Enterprise/RADIUS authentication was successful. Also, WLC Clients monitoring page is NOT supported in Packet Tracer

Evidence:



PC0

PhysicalConfigDesktopProgrammingAttributes

Web Browser

<>URLhttps://10.20.40.2/frameMonitor.htmlGoStop

CISCO

MONITORWLANsCONTROLLERWIRELESSSECURITYMANAGEMENTCOMMANDSHELPFEEDBACKHome

Monitor

Summary

Access Points

Cisco CleanAir

Statistics

CDP

Rogues

Clients

Sleeping Clients

Multicast

Applications

Local Profiling

Controller Summary

Management IP Address10.20.40.2, ::/128

Software Version8.3.111.0

Field Recovery Image Version7.6.101.1

System NameWLC1

Up Time23 minutes, 55 seconds

System TimeThu Oct 8 23:22:39 2026

Redundancy ModeN/A

Internal Temperature+31 C

802.11a Network StateEnabled

802.11b/g Network StateEnabled

Local Mobility Group

CPU(s) Usage0%

Individual CPU Usage0%/1%, 0%/0%

Memory Usage46%

Fan Status3800 rpm

Rogue Summary

Active Rogue APs0Detail

Active Rogue Clients0Detail

Adhoc Rogues0Detail

Rogues on Wired Network0

Top WLANs

Profile Name# of Clients

Most Recent Traps

View All

Top Applications

Application Name

Packet Count

Byte Count

View All

Access Point Summary

	Total	Up	Down	
802.11a/n/ac Radios	1	1	0	Detail
802.11b/g/n Radios	1	1	0	Detail
Dual-Band Radios	0	0	0	Detail
All APs	1	1	0	Detail

Client Summary

Current Clients0Detail

Excluded Clients0Detail

Disabled Clients0Detail

View All

☐ Top

RADIUS

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

AAA

Service

On

Off

Radius Port

1812

Network Configuration

Client NameClient IPSecretServerType

Radius

	Client Name	Client IP	Server Type	Key	
1	WLC1	10.20.40.2	Radius	Cisco123	Add
					Save
					Remove

User Setup

UsernamePassword

	Username	Password	
1	wifiuser	Cisco123	Add
			Save
			Remove

Top

Laptop0

Physical Config Desktop Programming Attributes

Creating a Profile

Wireless Security - WPA2 Enterprise

Authentication	PEAP	Please select the authentication method that you use to access your network.
Login Name	wifiuser	Enter the Login Name used for authentication.
Password	••••••••	Enter the Password used for authentication.
Server Name		Enter the Server Name used for authentication. (Optional)
Certificate	Trust Any	Please select the certificate used for authentication.
Inner Authen.	TOKEN CARD	Please select the inner authentication method used inside the PEAP tunnel.

BackNext

Wireless-N Notebook Adapter Wireless Network Monitor v1.11 Model No. WPC300N

Laptop0

Physical Config Desktop Programming Attributes

Link Information Connect Profiles

More Information Infrastructure Mode

You have successfully connected to the access point



Signal Strength  Link Quality  Adapter is Active

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. WPC300N

2.4GHz

To Do

PC0

PhysicalConfigDesktopProgrammingAttributes

Web Browser

<>URLhttps://10.20.40.2/frameMonitor.htmlGoStop

CISCO

MONITORWLANSCONTROLLERWIRELESSSECURITYMANAGEMENTCOMMANDSHelpFEEDBACKHome

Save Configuration | Ping | Logout | Refresh

Monitor

Summary

Access Points

Cisco CleanAir

Statistics

CDP

Rogues

Clients

Sleeping Clients

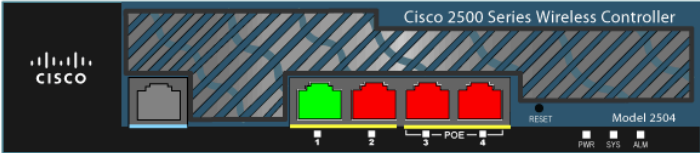
Multicast

Applications

Local Profiling

Summary

25 Access Points Supported



Controller Summary

Management IP Address

Software Version

Field Recovery Image Version

System Name

Up Time

System Time

Redundancy Mode

Internal Temperature +31 C

802.11a Network State Enabled

802.11b/g Network State Enabled

Local Mobility Group

CPU(s) Usage 0%

Individual CPU Usage 0%/1%, 0%/0%

Memory Usage 46%

Fan Status 3800 rpm

Rogue Summary

APs 0

Clients 0

802.11a 0

802.11b/g 0

Detail

Detail

Detail

Detail

AP Name

of Clients

Most Recent Traps

View All

Top Applications

Application Name

Packet Count

Byte Count

View All

Access Point Summary

	Total	Up	Down	
802.11a/n/ac Radios	1	1	0	Detail
802.11b/g/n Radios	1	1	0	Detail
Dual-Band Radios	0	0	0	Detail
All APs	1	1	0	Detail

JavaScript Alert - https://10.20.40.2/frameMonitor.html

This feature is not supported in Packet Tracer.

OK

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Task 2 – Syslog Capability Test

What was tested:

Packet Tracer was setup to test whether R1 could send log messages to a centralized Syslog server

Expected vs Actual Result:

The expected result was for the Syslog server to receive messages generated by R1. The server was able successfully receive log messages from 10.20.40.1

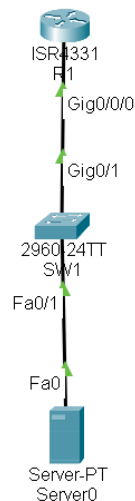
Result:

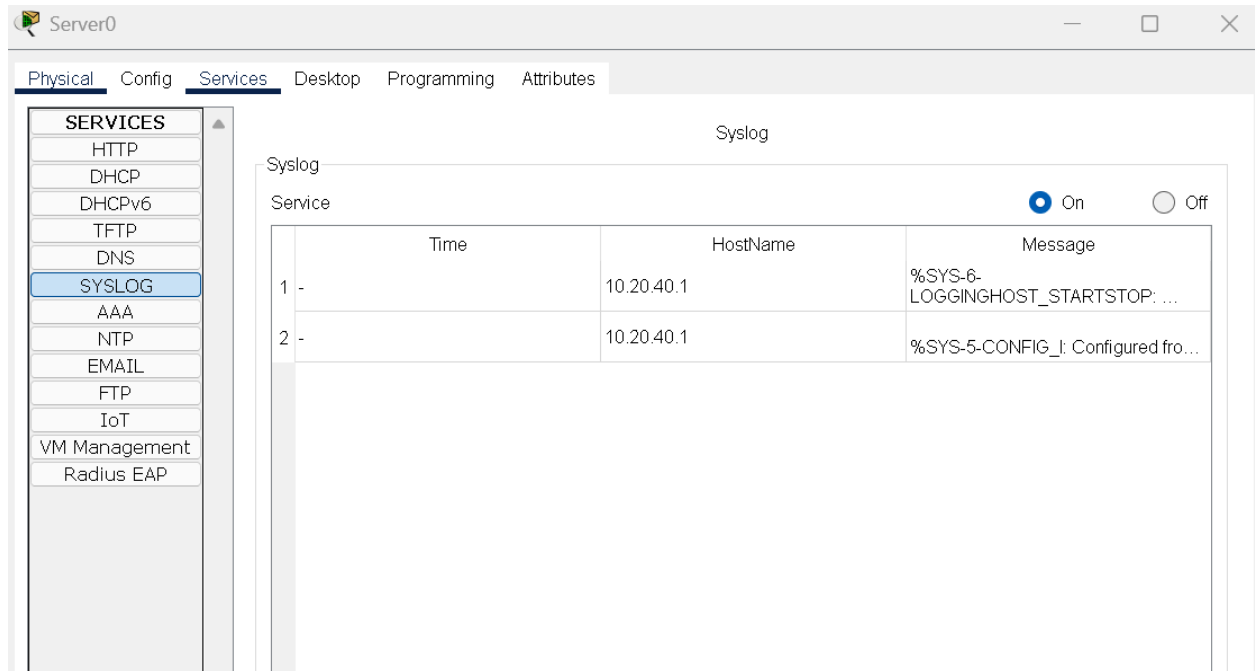
The Syslog capability test passed. Packet Tracer does support centralized Syslog logging from router to Server-PT

Limitation/Consideration:

Syslog Time field shows - because NTP has not been configured yet. Timestamp behavior will be checked during the NTP capability test.

Evidence:





Task 3 – NTP Capability Test

What was tested:

Packet tracer was used to test whether R1 could synchronize its clock with a centralized NTP server and use that time for syslog messages.

Expected vs Actual Result:

R1 was expected to synchronize with 10.20.40.12 and for the new Syslog messages to have timestamps included. R1 was successfully able to synchronize with the NTP server and there were new timestamps in Syslog messages.

Result:

The NTP capability test passed. Packet Tracer supports NTP synchronization and time stamped Syslog messages.

Limitation/Consideration:

Syslog entries usually showed - in the Time field. But when timestamps on R1 were enabled with "service timestamps log datetime msec", the new messages displayed the synchronized time.

Evidence:

The screenshot shows the configuration interface for a device named 'Server0'. The 'Services' tab is selected, and the 'SYSLOG' service is highlighted in the left-hand menu. The main area displays the 'Syslog' configuration, which is currently set to 'On'. Below this, a table lists the configured syslog services.

Service	Time	HostName	Message
1 -		10.20.40.1	%SYS-6-LOGGINGHOST_STARTSTOP: ...
2 -		10.20.40.1	%SYS-5-CONFIG_I: Configured fro...

R1

Physical

Config

CLI

Attributes

IOS Command Line Interface

Press RETURN to get started.

Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ntp server 10.20.40.12
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show clock
*4:0:48.293 UTC Mon Mar 1 1993
Router#show ntp associations

address	ref clock	st	when	poll	reach	delay	offset
disp							
~10.20.40.12	127.127.1.1	1	10	16	1	0.00	1060614649557.00
0.00							
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured							

Router#show clock
20:21:13.234 UTC Fri Oct 9 2026
Router#show ntp associations

address	ref clock	st	when	poll	reach	delay	offset
disp							
*~10.20.40.12	127.127.1.1	1	72	32	160	0.00	0.00
0.12							
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured							

Router#

Copy

Paste

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

Syslog

Service

On

Off

	Time	HostName	Message
1	-	10.20.40.1	%SYS-5-CONFIG_I: Configured from cons...
2	-	10.20.40.1	%SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.20.40.12 port 514 starte...
3	-	10.20.40.1	%SYS-5-CONFIG_I: Configured from cons...
4	10.09.2026 08:26:10.707 PM	10.20.40.1	%SYS-5-CONFIG_I: Configured from console by console

Clear Log

